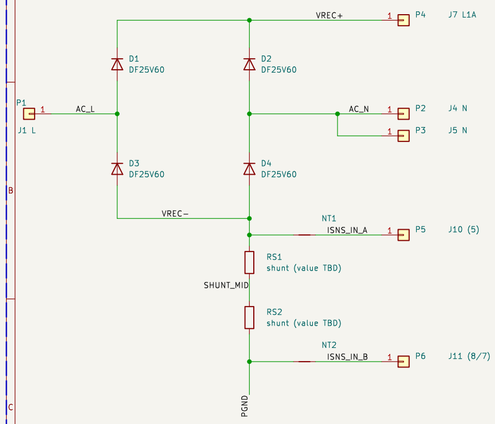
LeafOBC-WafflePlate: first-pass schematic of the 2012 Nissan Leaf OBC power module ("waffle plate")

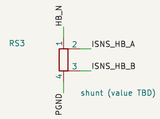
DRAFT. Sections A and D are meter-verified and drawn with wires; the rest uses net labels. Topology is from the community draft schematic plus identified part numbers. Connections are NOT yet verified against the module. Hidden fields per part: Verified, Draft_ref, Location_px (pixel position in images/stitched/waffle-plate-top.jpg).

A. Input bridge and shunts (upper left)



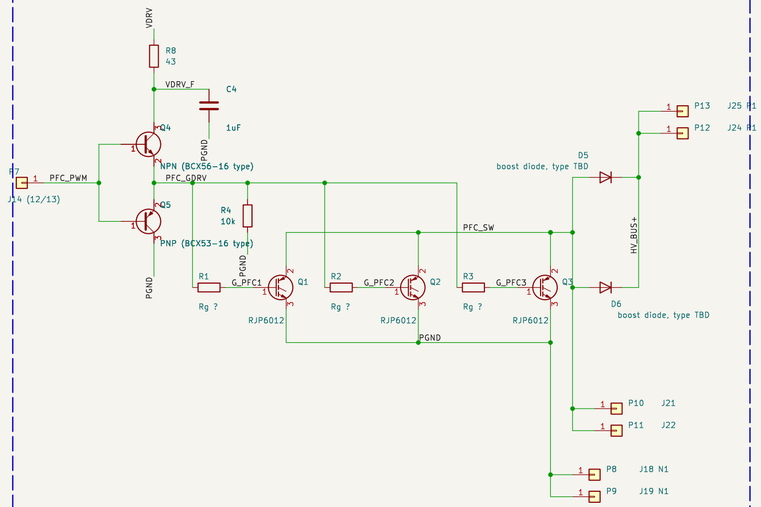
RS1 + RS2: one current shunt in two parts.

Kelvin sense taken from the outer terminals (NT1, NT2).



RS3: full-bridge return (section C)

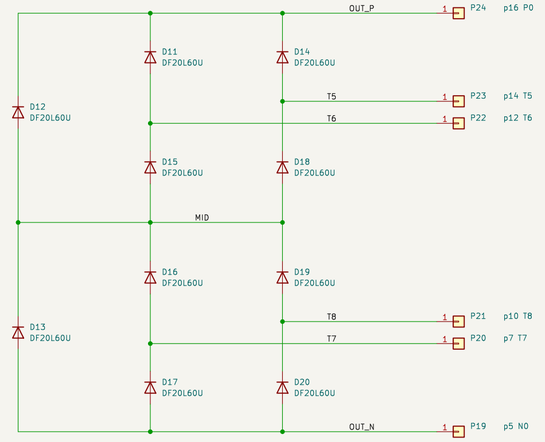
B. PFC boost stage (mid left)



Q1-Q3 in parallel, each with its own gate resistor. D5, D6 in parallel.

Gate driver (Q4, Q5, R4, R8, C4) meter-verified. IGBTs, gate resistors and boost diodes believed correct, not yet meter-verified.

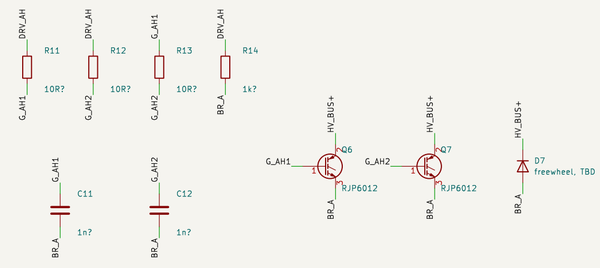
D. Output rectifier: two series-stacked bridges (upper right)



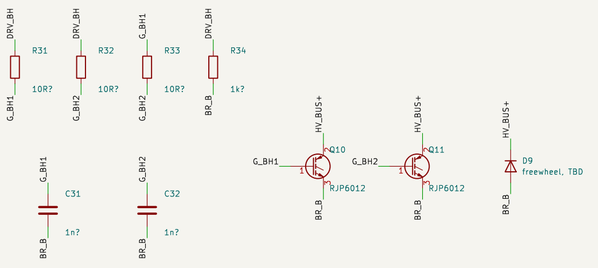
Verified with meter: D11-D20 and pads.

C. Primary full bridge: 4 switches, each 2 IGBTs in parallel + gate network (lower right)

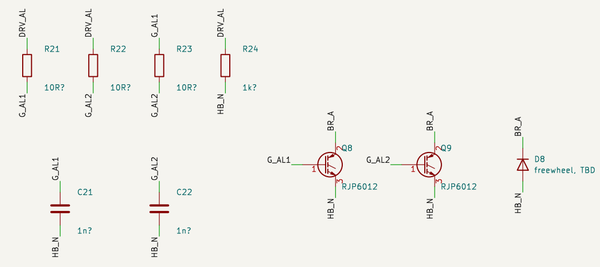
AH (by analogy)



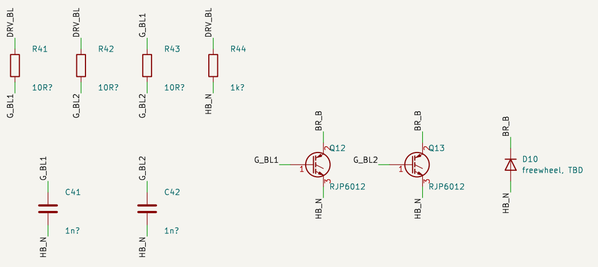
BH



AL (by analogy)

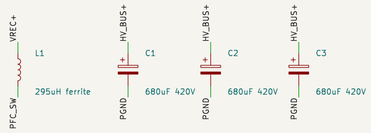


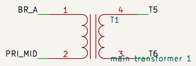
BL

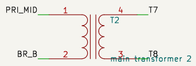


Values with ? are reference values (Ukrainian schematic and replica), unconfirmed. AL (Q8, Q9) failed on the reference module. See docs/full-bridge-gate-drive.md

E. Off-module parts (charger PCB)

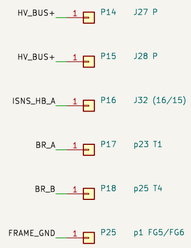






Primaries shown in series: verify.

F. Remaining module pads (draft numbering, unverified)



Pads for A, B and D are drawn next to their circuit.

Unassigned draft pads: J8 (6, IC312_PFC), J12/J13 (11), J15 (10/9), J30/J31 (N-N1), p27, p29, p31, p32, p33

CERN-OHL-S-2.0

Sections A and D meter-verified. Rest unverified.

LeafOBC-WafflePlate

Sheet: /

File: waffle-plate.kicad_sch

Title: Leaf OBC waffle plate: original module

Size: A2	Date:	**Rev: 0.1-draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1